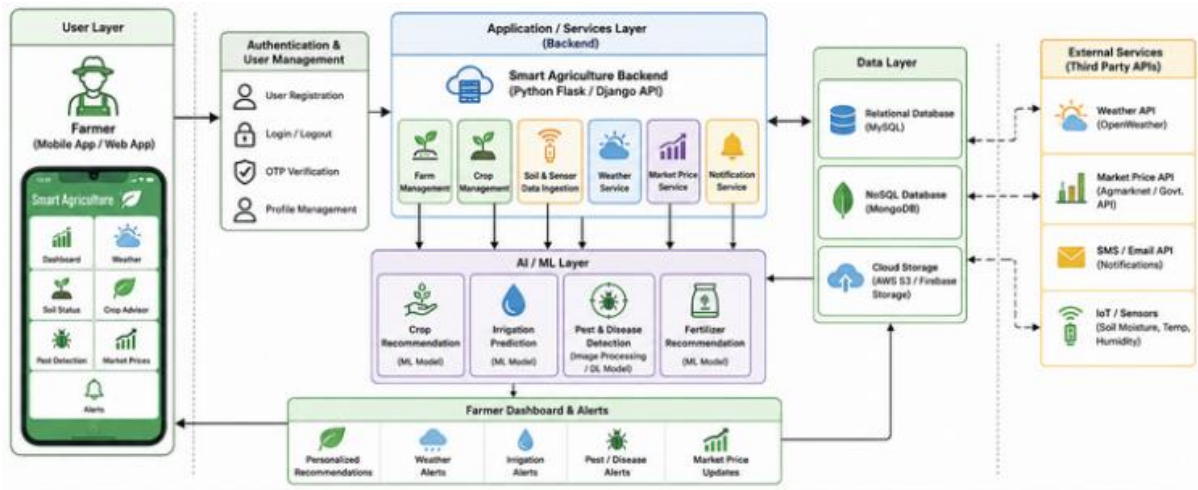


Project Design phase 2

Solution Requirements(functional & non functional)

Date	12 July 2026
Team id	
Project name	Smart Agriculture
Maximum marks	4 marks

Technical Architecture



Technical Components

S.No	Component	Description	Technology
1	User Interface	Mobile and Web application for farmers	Flutter / React.js, HTML, CSS, JavaScript
2	Authentication	User registration, login, OTP verification	Firebase Authentication
3	Application Logic-1	Farm management, crop details, user profile	Python (Flask/Django)
4	Application Logic-2	AI crop recommendation and irrigation prediction	Python, Scikit-learn, TensorFlow
5	Application Logic-3	Pest and disease detection using image processing	TensorFlow, OpenCV
6	Database	Stores farmer, crop, soil, and weather data	MySQL / MongoDB
7	Cloud Database	Cloud data storage and synchronization	Firebase Firestore / AWS RDS
8	File Storage	Crop images, reports, and documents	Firebase Storage / AWS S3
9	External API-1	Weather forecasting service	OpenWeather API

S.No	Component	Description	Technology
10	External API-2	Market price information	Agmarknet API / Government Agriculture API
11	Machine Learning Model	Predict crop suitability, irrigation schedule, pest detection	TensorFlow, Scikit-learn
12	Notification Service	Sends weather alerts and crop notifications	Firebase Cloud Messaging (FCM), Email API